

Student Name:	Priya Singh	Roll No:	102	BARCODE BAR003
Course:	B.Tech CSE	Semester:	IV	
Subject:	Mathematics — Mid-Term 2025	Barcode:	BAR003	

Answer to Q1(i)

Computer system is electronic device for data processing. Components are: 1. Input unit - keyboard mouse. 2. CPU - processes instructions. 3. Memory - RAM stores temporary data. 4. Output - monitor printer shows results. CPU is brain of computer containing ALU and Control Unit.

—
—
—
—

Answer to Q1(ii)

Primary memory is RAM. It is volatile, fast, costly. Used by CPU directly. Secondary memory is hard disk. Non-volatile, slow, cheap, large capacity. RAM examples: 8GB DDR4. HDD examples: 1TB Seagate drive.

—
—
—
—

Answer to Q1(iii)

Operating system manages hardware and provides interface to user. Functions: Memory management distributes RAM among programs. Process management handles CPU scheduling. File system manages data storage. I/O management controls devices. User interface - GUI or CLI for user interaction.

—
—
—
—

BARCODE
BAR003**Answer to Q2(i)**

Von Neumann model: CPU contains CU + ALU + Registers. Main memory stores program and data. Input devices send data in. Output devices show results. System bus connects all parts. This is stored program concept.

—
—
—
—

Answer to Q3(i)

255 decimal to binary: $255/2=127$ r1, $127/2=63$ r1, $63/2=31$ r1, $31/2=15$ r1, $15/2=7$ r1, $7/2=3$ r1, $3/2=1$ r1, $1/2=0$ r1. Binary = 11111111. Octal = 377. Hex = FF.

—
—
—
—

Answer to Q4(i)

Computers by size: Supercomputer - for scientific research, fastest. Mainframe - for large organisations, banking. Minicomputer - mid size server. Microcomputer - personal computer, laptop. Embedded - ATM machines. By purpose: General purpose - home use. Special purpose - dedicated tasks.

—
—
—
—

Answer to Q7(i)

SRAM uses flip-flops. Faster, expensive, no refresh, used in cache. DRAM uses capacitors, needs refresh every few milliseconds, slower, cheaper. DRAM used in main memory. SRAM in L1, L2 cache. DRAM has higher density.

—
—
—
—